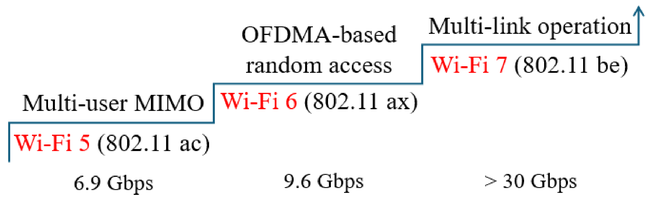
# Saturation Throughput Analysis of UORA-Enabled Multi-Link Operation in IEEE 802.11be Wi-Fi

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## 1. Wi-Fi Evolution: Features & Challenges

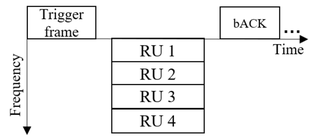
- Wi-Fi connects billions of devices worldwide
  - Significant portion of cellular traffic offloaded to Wi-Fi
- Evolved significantly

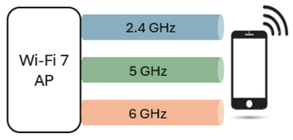



Operates in unlicensed bands and uses contention-based access

## 2. Key Features in Wi-Fi 7 (IEEE 802.11be)

- Uplink OFDMA random access (UORA)
  - Enable concurrent transmissions
  - AP controls UORA via trigger frames



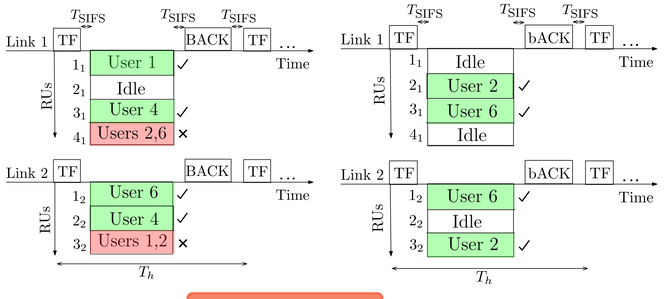

- Multi-link operation (MLO)
  - Device can transmit and receive on multiple radio links
  - Enables higher throughput & lower delay

## 3. Multi-link Operation with UORA

Non-simultaneous MLO supports transmit-transmit and receive-receive
- Avoids more sophisticated hardware

- Two possible back-off rules: device transmits on both links if
  - *Shortest back-off rule:* counter of any link is zero
  - *Longest back-off rule:* only if counters of both links are zero

| User ID | Counter | |
|---|---|---|
| | Link 1 | Link 2 |
| 1 | 0 | 4 |
| 2 | 0 | 0 |
| 3 | 2 | 19 |
| 4 | 31 | 0 |
| 5 | 1 | 7 |
| 6 | 0 | 0 |

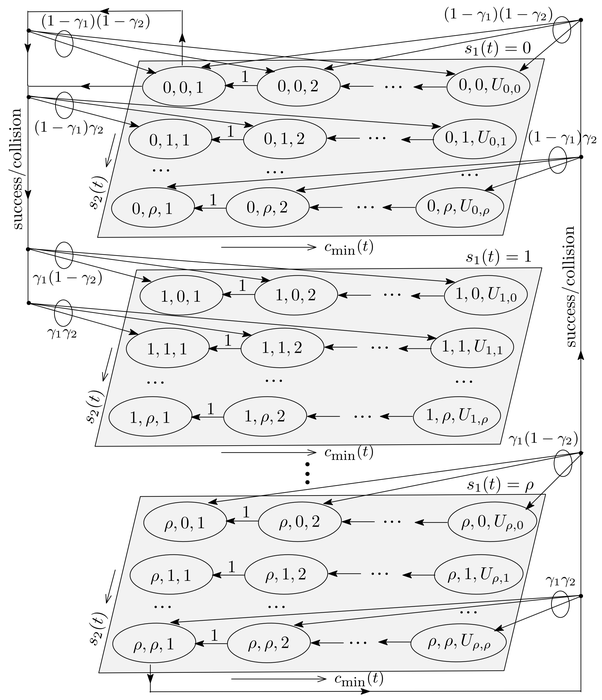



**Contributions**
- Characterize the joint evolution of back-off processes of two links
- Throughput analysis of non-simultaneous MLO with UORA

## 4. Analytical Approach

- Attempt and collision processes are coupled → $O(K^{\rho^2})$ states
  - $K$: Number of users and $\rho = \log_2(C_{\max}/C_{\min})$
- Employ decoupling approximations [Bianchi 2000, Kumar 2007]

Shortest back-off rule state: $(s_1(t), s_2(t), c_{\min}(t))$
$s_l(t)$: link $l$ back-off stage, $c_{\min}(t)$: minimum back-off duration




- Attempt probability: $\beta = \sum_{i=0}^{\rho}\sum_{j=0}^{\rho} b_{i,j,1}$
  - $b_{i,j,k} = \lim_{t\to\infty} \Pr\left((s_1(t)=i, s_2(t)=j, c_{\min}(t)=k)\right)$
- Collision probability: $\gamma_l = 1-(1-(\beta/N_l))^{K-1}$

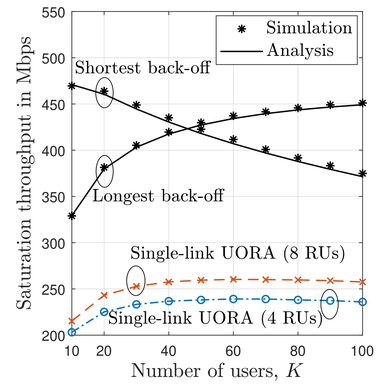
## 5. Saturation Throughput

Throughput on link $l$: $\Theta_l = \frac{T_P \bar{R}_l}{T_h} K\beta(1-\gamma_l)$

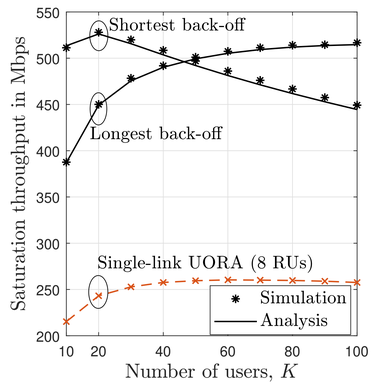
- $T_P$: payload duration
- $R_l$: average rate of a user with rate adaptation

$$\bar{R}_l = \frac{2d_0^2(P_{l,\text{tx}}c^2)^{\frac{2}{\eta}}}{\eta(d_{\max}^2-d_{\min}^2)(4\pi d_0 f_l \sigma)^{\frac{4}{\eta}}} \sum_{m\in\Delta} \frac{r_{l,m}-r_{l,m-1}}{\Upsilon_m^{\frac{2}{\eta}}} \times \left[\gamma_{\text{inc}}\left(2/\eta, \lambda_{l,m} d_{\max}^{\eta}\right) - \gamma_{\text{inc}}\left(2/\eta, \lambda_{l,m} d_{\min}^{\eta}\right)\right]$$

## 6. Throughput of Multi-link with UORA



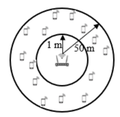

(8 RUs in link 1, 4 RUs in link 2)



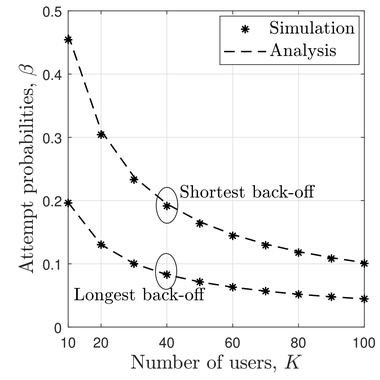

(8 RUs in link 1, 8 RUs in link 2)

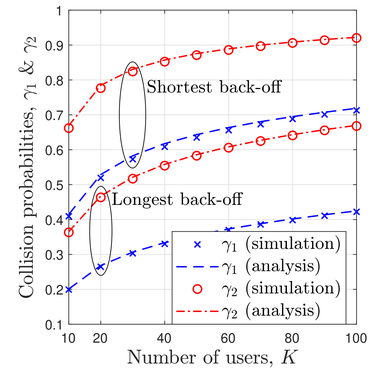
(160 MHz BW, $T_P = 5$ ms, $C_{\min}=16$, $C_{\max}=1024$)

- Shortest back-off has higher throughput for small $K$
- Longest back-off has higher throughput for larger $K$




## 7. Inside the Workings: Back-off Parameters




(8 RUs in link 1, 4 RUs in link 2)

- Attempt probability of the shortest rule is higher than the longest rule
- Collision probability depends on the number of UORA RUs

## 8. Key Takeaways and Extensions

- Multi-link UORA has a higher throughput than single-link UORA
- Throughput gains depend on the number of RUs and back-off rule

**Extensions**
- Throughput and delay of MLO + UORA with legacy users
- Performance with unsaturated traffic